Case Study 9: Student Marks Matrix Analysis

Assignment Case No. 42 | Level: Beginner

01 PROBLEM STATEMENT

Analyse a matrix of student marks across subjects, calculate row-wise and column-wise statistics, identify top performers, and detect subjects with low average performance.

Problem interpretation: Marks are stored in a nested list where each row represents a student and each column represents a subject. The program calculates student totals and subject averages, then identifies the top student and subjects with low averages.

02 OBJECTIVES

Represent marks using a nested list.

Calculate each student's total and average.

Calculate average marks for each subject.

Identify the top performer and subjects with averages below a threshold.

03 ASSUMPTIONS

Rows represent students and columns represent subjects in the same order.

Marks are between 0 and 100.

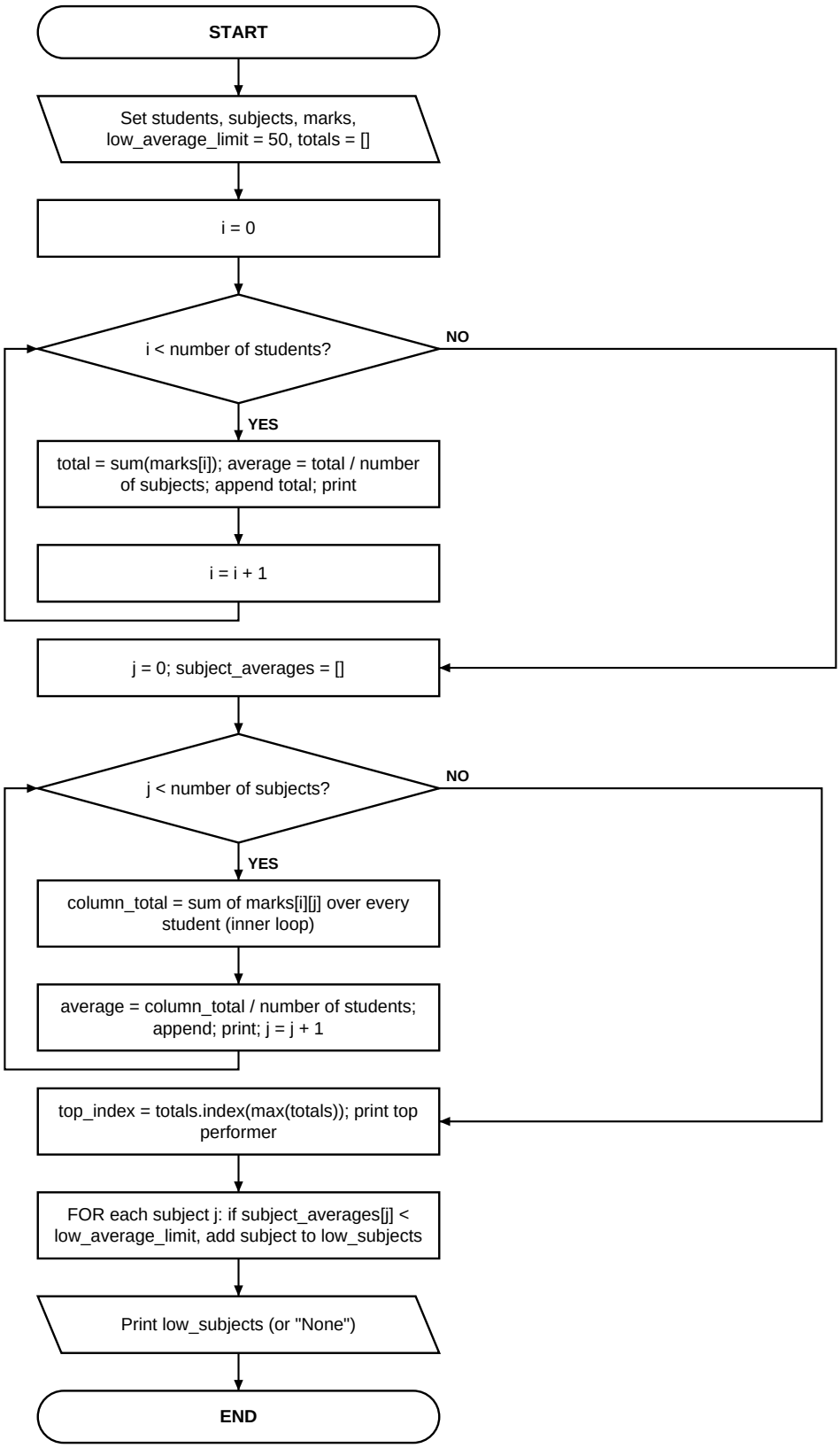
The subject-average warning threshold is 50.

The example contains at least one student and one subject.

04 ALGORITHM

1. Start.
2. Store student names, subject names, and the marks matrix.
3. For each student, calculate the row total and average.
4. For each subject, sum the column and divide by student count.
5. Find the student with the highest total.
6. Flag subjects with average below 50.
7. Display all statistics.
8. Stop.

05 FLOWCHART (TEXT FLOW)



06 PSEUDOCODE

```
BEGIN
    students = [Asha, Ravi, Noor]
    subjects = [Python, Maths, English]
    marks = [[80, 75, 90], [65, 85, 70], [45, 55, 60]]
    low_average_limit = 50; totals = []
    FOR each student i
        total = sum(marks[i])
        average = total / number of subjects
        add total to totals
        PRINT student, total, average
    END FOR
    subject_averages = []
    FOR each subject j
        column_total = 0
        FOR each student i
            column_total = column_total + marks[i][j]
        END FOR
        average = column_total / number of students
        add average to subject_averages
        PRINT subject, average
    END FOR
    top_index = position of maximum in totals
    PRINT top performer
    low_subjects = []
    FOR each subject j
        IF subject_averages[j] < low_average_limit THEN add subjects[j] to low_subjects
    END FOR
    PRINT low_subjects, or "None" if empty
END
```

07 PYTHON SOURCE CODE

The complete Python program is provided in the separate file 09_Student_Marks_Matrix_Analysis_Code.pdf.

08 TEST CASES

INPUT / SCENARIO	CHECK	EXPECTED RESULT
Asha: 80, 75, 90	Total 245; average 81.67	Top performer
Python column: 80, 65, 45	Average 63.33	Not flagged
Sample data	All subject averages above 50	Subjects below threshold: None
Python column: 45, 45, 45	Average 45, below 50	Python flagged

09 SAMPLE OUTPUT PREVIEW

```
STUDENT TOTALS AND AVERAGES
Asha: total=245, average=81.67
Ravi: total=220, average=73.33
Noor: total=160, average=53.33

SUBJECT AVERAGES
Python: 63.33
Maths: 71.67
English: 73.33
Top performer: Asha (245 marks)
Subjects below average threshold: None
```

10 CONCLUSION

The program uses nested lists and nested loops to calculate row-wise and column-wise statistics and identify the top performer.